

Discussion Outline – Nutrient Economics and iWUE in Tropical Montane Forests

Core message: Long-term fertilization altered nutrient economics but not carbon–water economics.

Introduction paragraph

After 18 years of nutrient addition, foliar nutrient concentrations increased, but iWUE remained largely unchanged. This suggests a decoupling between nutrient acquisition and carbon–water regulation in mature tropical montane forest trees.

4.1 Long-term fertilization altered foliar nutrient status

Question: Did leaves respond to fertilization? Yes.

Main ideas: (1) Experiment successfully increased LNC, LPC and Narea in some species; (2) evidence supports phosphorus limitation; (3) persistent nutrient uptake may reflect adaptation to nutrient-poor environments.

4.2 Changes in nutrient concentrations did not translate into higher iWUE

Question: Why did nutrients increase but iWUE remain unchanged?

Main ideas: (1) iWUE, $\delta^{13}\text{C}$ and Ci/Ca showed weak responses; (2) photosynthetic economics predicts stronger responses, but expectations were not met; (3) nutrient acquisition alone may be insufficient to alter carbon–water relations.

4.3 Nutrient economics and allocation strategies may explain the weak iWUE response

Question: Where did the additional nutrients go?

Main ideas: (1) Total N is not equivalent to photosynthetic N; (2) Wright et al. (2023) and nutrient economics theory suggest nutrients may be allocated to storage, structure, defense, canopy development, or acquisition of other limiting resources; (3) increased LAI and leaf longevity indicate structural rather than physiological responses; (4) conservative species strategies may reinforce these allocation patterns.

4.4 Elevation exerts a stronger control on iWUE than nutrient availability

Question: If nutrients do not explain iWUE, what does?

Main ideas: (1) Higher iWUE at mid and high elevations; (2) consistency with Plant Economics Spectrum; (3) optimality theory predicts higher iWUE under lower atmospheric CO_2 partial pressure; (4) peak iWUE around 2000 m may be linked to elevated photosynthetic capacity.

4.5 Conclusions and implications

Nutrient enrichment clearly modified foliar nutrient status, but not iWUE. Nutrients may have been allocated to storage, retention, and canopy development rather than directly increasing carbon assimilation. Elevation appears to be a stronger determinant of carbon–water economics than nutrient availability.